

Paper 2 - Set 12 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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1	F	Straight lines
2	D	Polynomials, factor and remainder theorems
3	C	Quadratic functions
4	A	Identifying errors in proofs
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Worked solutions

Question 1 LIN-02 · Straight lines · spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

$$\text{Gradient } AB = \frac{2}{4} = \frac{1}{2}. \text{ Gradient } AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k-1 = 12 \Rightarrow k = 13.$$

Full method - how and why

Collinear points have equal gradients between any pair. Set two gradients equal and solve.

1 Gradient of AB : $\frac{4-2}{5-1} = \frac{2}{4} = \frac{1}{2}$.

why: Any two of the three points determine the line, so compute the gradient from the pair with known coordinates.

2 Gradient of AC : $\frac{8-2}{k-1} = \frac{6}{k-1}$.

why: Use the same starting point A for both gradients to avoid sign confusion.

3 Set them equal: $\frac{6}{k-1} = \frac{1}{2}$.

why: Collinear means the gradient between any pair is the same.

4 Cross-multiply: $k-1 = 12$, so $k = 13$.

why: Check: from $B(5, 4)$ to $C(13, 8)$ the gradient is $\frac{4}{8} = \frac{1}{2}$ ✓.

Where it goes wrong

- Forgetting the -1 and answering 12.
- Inverting the gradient and solving $\frac{k-1}{6} = \frac{1}{2}$, which gives $k = 4$.

Takeaway Collinearity is an equal-gradient condition; always verify with the third pair of points.

Question 2 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$$p(1) = 1 + a + b + c, \text{ and } p(1) = (1-2)(1+1)(1-3) = (-1)(2)(-2) = 4. \text{ So } a + b + c = 3. \text{ (Directly: } (x-2)(x+1)(x-3) = x^3 - 4x^2 + x + 6.)$$

Full method - how and why

You could expand, but $a + b + c$ is the sum of the non-leading coefficients, which $p(1)$ delivers instantly.

1 Note $p(1) = 1 + a + b + c$, so $a + b + c = p(1) - 1$.

why: Substituting $x = 1$ turns every power into 1, so $p(1)$ is the sum of **all** coefficients including the leading 1.

2 Write the factorised form from the roots: $p(x) = (x-2)(x+1)(x-3)$.

why: Monic, because the given polynomial starts with x^3 .

3 Evaluate: $p(1) = (-1)(2)(-2) = 4$.

why: Two negatives and one positive give a positive.

4 So $a + b + c = 4 - 1 = 3$.

why: Remember to subtract the leading coefficient.

Where it goes wrong

- Answering 4 by forgetting the leading 1 in $p(1) = 1 + a + b + c$.
- Expanding and mis-collecting; the expansion is $x^3 - 4x^2 + x + 6$, so $-4 + 1 + 6 = 3$ ✓.

Takeaway $p(1)$ is the sum of all coefficients - a one-second route to questions like this.

Question 3 QUA-24 · Quadratic functions · spec MM1.3

Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

Answer: C

$(x - 4)(x + 2) < 0$ gives $-2 < x < 4$; the integers are $-1, 0, 1, 2, 3$: five of them.

Full method - how and why

Counting **integers** in the region where a quadratic is negative. Solve the inequality, then count carefully at the ends.

1 Factorise: $n^2 - 2n - 8 = (n - 4)(n + 2)$.

why: Two numbers multiplying to -8 and adding to -2 : -4 and $+2$.

2 Negative between the roots: $-2 < n < 4$.

why: Upward parabola, so $f < 0$ strictly between the roots.

3 List the integers strictly inside: $-1, 0, 1, 2, 3$ - five of them.

why: The endpoints -2 and 4 give $f = 0$, which is **not** less than zero, so they are excluded; and 0 is easy to forget.

Where it goes wrong

- Including -2 and 4 and answering 7 .
- Forgetting 0 is an integer and answering 4 .

Takeaway After solving an inequality, list the integers explicitly - endpoints and zero are the two things people drop.

Question 4 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

- I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.
- II So $x^2 - 7x + 6 = 0$.
- III So $(x-1)(x-6) = 0$.
- IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Full method - how and why

Squaring an equation is not reversible: it can create solutions that do not satisfy the original. The error is the **missing check**, not the algebra.

- 1 Check line I: squaring both sides of $\sqrt{x+3} = x-3$ gives $x+3 = x^2 - 6x + 9$. Valid.
why: $A = B \Rightarrow A^2 = B^2$ is always a legitimate deduction - it is the converse that fails.
- 2 Check lines II and III: $x^2 - 7x + 6 = 0$ factorises as $(x-1)(x-6) = 0$. Valid.
why: $-1 \times -6 = 6$ and $-1 - 6 = -7$ ✓.
- 3 Line IV asserts both are solutions. Substitute $x = 1$ into the **original**: $\sqrt{4} = 2$, but $x-3 = -2$. $2 \neq -2$, so $x = 1$ is extraneous.
why: A square root is non-negative, so the right-hand side must be ≥ 0 , forcing $x \geq 3$.
- 4 Substitute $x = 6$: $\sqrt{9} = 3$ and $6-3 = 3$ ✓. So only $x = 6$ is a solution, and the first error is line IV.
why: Lines I-III produce **candidates**; the check is a required final step.

Where it goes wrong

- Blaming line I - squaring is a valid implication, just not an equivalence.
- Declaring the solution correct because the factorising is right; correct algebra can still give a wrong answer set.
- Checking the candidates in the squared equation rather than the original - both pass there, which proves nothing.

Takeaway After squaring, always substitute back into the **original** equation; squaring can only add solutions, never lose them.

Question 5 X2-17 · Integration · spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Full method - how and why

Whether the trapezium rule over- or under-estimates depends on the **concavity** of the curve.

- 1 The trapezium rule replaces the curve on each strip by the straight chord joining the endpoints.

why: Its estimate is the area under those chords.

- 2 $y = \sqrt{x}$ is concave: it rises steeply then flattens, bending downwards.

why: Its second derivative $-\frac{1}{4}x^{-3/2}$ is negative for $x > 0$ - or just picture the shape.

- 3 For a concave curve every chord lies **below** the arc, so each trapezium sits below the true region.

why: That is what concavity means geometrically.

- 4 So the estimate is an underestimate, for every n .

why: The number of strips affects accuracy, not direction.

Where it goes wrong

- Assuming the trapezium rule always overestimates - true only for convex (upward-bending) curves such as $y = x^2$.
- Thinking the parity of n matters - that is Simpson's rule, which needs an even number of strips.
- Answering 'cannot be decided' when the shape settles it.

Takeaway Trapezium rule: convex curve $\Rightarrow$ overestimate, concave curve $\Rightarrow$ underestimate. Sketch and look at the bend.

Question 6 PRF-14 · Mathematical proof · spec Prf1

To prove '**if n^2 is even then n is even**' by contradiction, which assumption should be made?

Answer: B

Assume the hypothesis holds and the conclusion fails, then derive a contradiction: n odd gives n^2 odd, contradicting n^2 even.

Full method - how and why

Setting up a proof by contradiction for an implication: assume the hypothesis **and** the negated conclusion.

- 1 The claim is 'if n^2 is even then n is even': hypothesis n^2 even, conclusion n even.

why: Identify both parts before negating anything.

- 2 For contradiction, suppose the whole implication fails. An implication fails only when the hypothesis holds and the conclusion does not.

why: This is the one case that would make $A \Rightarrow B$ false.

- 3 So assume: n^2 is even **and** n is odd.

why: Both, not just one.

- 4 (The proof then runs: $n = 2k + 1 \Rightarrow n^2 = 4k^2 + 4k + 1$, odd - contradicting n^2 even.)

why: The contradiction is with the retained hypothesis.

Where it goes wrong

- Assuming only ' n is odd' and dropping the hypothesis - then there is nothing to contradict.
- Assuming ' n^2 is odd', which negates the hypothesis instead of the conclusion.
- Assuming ' n even and n^2 odd', which negates the wrong part of each.

Takeaway To contradict $A \Rightarrow B$, assume A **and** $\neg B$ - keep the hypothesis, negate only the conclusion.

Question 7 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.

Full method - how and why

A quadrilateral hierarchy question. Bisecting diagonals characterise parallelograms, a larger class than rectangles.

- 1 Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other. So $Q \Rightarrow P$: P is necessary.
why: Opposite sides equal and parallel forces the diagonals to share a midpoint.
- 2 Is P sufficient? Take a non-rectangular parallelogram, e.g. a rhombus with angles $60^\circ/120^\circ$: diagonals bisect each other, but it is not a rectangle.
why: Bisecting diagonals say nothing about the angles.
- 3 So P is necessary but not sufficient.
why: For a rectangle you also need the diagonals to be **equal** in length.

Where it goes wrong

- Answering 'necessary and sufficient' by confusing 'bisect each other' with 'bisect each other and are equal'.
- Getting the direction backwards: the broader property (parallelogram) is necessary for the narrower one (rectangle).

Takeaway Diagonals bisect each other $\Leftrightarrow$ parallelogram; add equal diagonals for a rectangle, add perpendicular for a rhombus.

Question 8 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement $\forall x$ 'every student in the class scored more than 50'?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Full method - how and why

Negating a universal statement. 'All' flips to 'at least one', and the property inside flips too.

- 1 Write it formally: 'for all students s , $\text{mark}(s) > 50$ '.
why: Formalising stops the English from misleading you.
- 2 Negate: 'there exists a student s with $\text{mark}(s) \not> 50$ '.
why: $\neg \forall x P(x) \equiv \exists x \neg P(x)$: the quantifier flips and the inside is negated.
- 3 Translate $\not> 50$ into ' ≤ 50 ', i.e. ≤ 50 or less.
why: The negation of 'greater than' is 'less than **or equal to**' - the boundary belongs to the negation.
- 4 So: at least one student scored 50 or less.
why: One failure is all it takes to make 'every' false.

Where it goes wrong

- Over-negating to 'every student scored 50 or less' - that is far stronger than the negation and is not implied.
- Writing 'less than 50' and losing the case of exactly 50.

Takeaway Negating 'all ... are' gives 'at least one ... is not' - and remember $\neg(x > a)$ is $x \leq a$.

Question 9 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x - 2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x - 2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Full method - how and why

The factor theorem gives one direction; a repeated factor is strictly more than a root.

- 1 If $(x - 2)^2$ is a factor, then $(x - 2)$ is a factor, so $p(2) = 0$: $Q \Rightarrow P$. **** P is necessary.****
why: Factor theorem: $(x - a)$ is a factor exactly when $p(a) = 0$.
- 2 Is P sufficient? Take $p(x) = x - 2$: $p(2) = 0$, but $(x - 2)^2$ does not divide a degree-1 polynomial.
why: A single root is not a repeated root.
- 3 So P is necessary but not sufficient.
why: Note the direction: the weaker condition is necessary for the stronger one.

Where it goes wrong

- Answering 'sufficient but not necessary' - the common pattern, but reversed here.
- Thinking $p(2) = 0$ and $p'(2) = 0$ together would be needed; that is exactly the extra condition for a repeated root, and its absence is the point.

Takeaway $p(a) = 0 \Leftrightarrow (x - a)$ is a factor; a **repeated** factor also needs $p'(a) = 0$.

Question 10 CIR-11 · Circles and circle theorems · spec MM3.2

Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point $(5, 0)$ lies on both circles.

Answer: F

Radii 5 and 3, centres 8 apart = $5 + 3$, so the circles touch externally at one point (I true, II false). That point is $(5, 0)$: $25 = 25$ and $9 + 0 = 9$ (III true).

Full method - how and why

Two circles: compare the distance between centres with the sum and difference of the radii.

- 1 Read off centres and radii: $(0, 0)$ with $r = 5$; $(8, 0)$ with $r = 3$.
why: Both are already in centre-radius form.
- 2 Distance between centres: 8. Compare with $r_1 + r_2 = 8$.
why: Equal to the sum means the circles touch externally - exactly one common point.
- 3 I is therefore true, and II (two intersection points) is false.
why: Two intersections would need $|r_1 - r_2| < d < r_1 + r_2$, i.e. $2 < d < 8$ strictly.
- 4 III: check $(5, 0)$ in both: $25 + 0 = 25$ ✓ and $(5 - 8)^2 + 0 = 9$ ✓. **True.**
why: The touching point lies on the line joining the centres, 5 from one and 3 from the other.
- 5 I and III only.
why: Match to the options.

Where it goes wrong

- Assuming circles that 'meet' must meet twice.
- Forgetting to test III in **both** equations.

Takeaway $d = r_1 + r_2$: touch externally; $d = |r_1 - r_2|$: touch internally; strictly between: two points.

Question 11 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Full method - how and why

Inclusion-exclusion: count what you don't want and subtract. Take care with 'less than 1000'.

- 1 The range is 1 to 999, so 999 integers.
why: 'Less than 1000' excludes 1000 itself.
- 2 Multiples of 2: $\lfloor \frac{999}{2} \rfloor = 499$. Multiples of 5: 199. Multiples of both, i.e. of 10: 99.
why: Divide and round down. 10 is the LCM of 2 and 5.
- 3 Divisible by 2 or 5: $499 + 199 - 99 = 599$.
why: Subtract the overlap, which was counted twice.
- 4 Neither: $999 - 599 = 400$.
why: Sanity check: in each block of 10 consecutive integers exactly 4 (ending in 1, 3, 7, 9) survive, and $999 = 99$ full blocks plus 991-999, giving $99 \times 4 + 4 = 400$ ✓.

Where it goes wrong

- Using 1000 as the count and getting 400 by luck with wrong intermediate figures - or getting 399 by an off-by-one at the top.
- Forgetting to subtract the multiples of 10, giving $999 - 698 = 301$.

Takeaway $|A \cup B| = |A| + |B| - |A \cap B|$, with the intersection counted via the LCM; then subtract from the total.

Question 12 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

- I** $\sin^2 x + \cos^2 x = 1$
II $\sin(2x) = 2\sin x$
III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$: $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.

Full method - how and why

Three identity claims. Two are standard, one is a classic false 'linearity' claim.

- 1 I: $\sin^2 x + \cos^2 x = 1$ is the Pythagorean identity, true for every x . **True.**
why: It follows from Pythagoras applied to the unit circle.
- 2 II: test $x = \frac{\pi}{2}$: $\sin \pi = 0$ but $2\sin \frac{\pi}{2} = 2$. **False.**
why: Trig functions are not linear: $\sin(2x) \neq 2\sin x$ (the correct identity, off-specification here, is $2\sin x \cos x$).
- 3 III: shifting $\sin$ by half a period reflects it: $\sin(x + \pi) = -\sin x$. **True.**
why: Check at $x = 0$: $\sin \pi = 0 = -\sin 0$ ✓; at $x = \frac{\pi}{2}$: $\sin \frac{3\pi}{2} = -1 = -\sin \frac{\pi}{2}$ ✓.
- 4 I and III only.
why: Match to the options.

Where it goes wrong

- Accepting II because it 'looks like' a factor coming out.
- Doubting III - test it at two values to be sure.

Takeaway Functions are almost never linear: $f(2x) \neq 2f(x)$ in general. Test suspicious identities at $x = \frac{\pi}{2}$.

Question 13 DIF-11 · Differentiation · spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Full method - how and why

Three claims about a cubic. The derivative settles the first two; the third is about global behaviour, where cubics are unbounded.

- Differentiate: $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$.
why: The roots ± 1 are the stationary points.
- I: for $x > 1$ both factors are positive, so $f' > 0$ and f is increasing. **True.**
why: Sign analysis of the factorised derivative.
- II: $f'' = 6x$, so $f''(-1) = -6 < 0$: local maximum at $x = -1$. **True.**
why: Negative second derivative means a local maximum.
- III: the local minimum is $f(1) = -2$, but as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ - for instance $f(-3) = -27 + 9 = -18$. **False.**
why: A **local** minimum is not a global one; odd-degree polynomials are unbounded below.
- I and II only.
why: Match to the options.

Where it goes wrong

- Assuming the local minimum value bounds the whole function.
- Confusing the maximum and minimum positions - for $+x^3$ the maximum is at the smaller x .

Takeaway Cubics have no global maximum or minimum; 'local' claims need f' , 'global' claims need the end behaviour.

Question 14 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x-y)(x+y) = 45$ with $0 < x-y < x+y$: (1, 45), (3, 15), (5, 9) give $(x, y) = (23, 22), (9, 6), (7, 2)$. Three pairs.

Full method - how and why

A difference of two squares turns a quadratic condition into a factor-pair search.

- Factorise: $x^2 - y^2 = (x-y)(x+y) = 45$.
why: Difference of two squares converts the equation into a product of two integers.
- Both factors are positive with $x-y < x+y$, so list factor pairs of 45 with the smaller first: (1, 45), (3, 15), (5, 9).
why: x, y positive forces $x+y > 0$, and then $x-y > 0$ too since the product is positive.
- Solve each: $x = \frac{\text{sum}}{2}$, $y = \frac{\text{difference}}{2}$, giving (23, 22), (9, 6), (7, 2).
why: Adding the two equations gives $2x$; subtracting gives $2y$. Both factors are odd, so both results are integers.
- All three have positive integer y , so there are **three** pairs.
why: Check one: $9^2 - 6^2 = 81 - 36 = 45$ ✓.

Where it goes wrong

- Including (9, 5) or other reversed pairs - $x-y$ must be the smaller factor.
- Forgetting to check that x and y come out as integers; if 45 had a factor pair of mixed parity it would have to be discarded.
- Counting $45 = 45 \times 1$ and 1×45 as two different solutions.

Takeaway $x^2 - y^2 = N$: list factor pairs $N = de$ with $d < e$ and the same parity, then $x = \frac{d+e}{2}$, $y = \frac{e-d}{2}$.

Question 15 PRF-11 · Mathematical proof · spec Prf3

A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?

Answer: A

Line 1 starts at 1, line 2 at 4, line 3 at 9: start n^2 . Left sides have 2, 3, 4 numbers ($n + 1$), right sides 1, 2, 3 (n).

Full method - how and why

Describing a pattern precisely: check the start value and the two counts separately against every given line.

- 1 Starting values: 1, 4, 9 for lines 1, 2, 3 - these are n^2 .

why: Test against all three lines, not just one.

- 2 Count the left sides: {1, 2}, {4, 5, 6}, {9, 10, 11, 12} - that is 2, 3, 4 terms, i.e. $n + 1$.

why: Count carefully; being off by one here is the whole point of the question.

- 3 Count the right sides: {3}, {7, 8}, {13, 14, 15} - that is 1, 2, 3 terms, i.e. n .

why: Left has one more term than right, which is what makes the equality possible (the right-hand terms are larger).

- 4 So: starts at n^2 , with $n + 1$ terms on the left and n on the right.

why: Check the total count: $(n + 1) + n = 2n + 1$, and indeed line 3 runs from 9 to 15, seven numbers ✓.

Where it goes wrong

- Swapping the counts (n left, $n + 1$ right) - the sides would not balance, since the right-hand numbers are bigger.
- Reading the start as $2n - 1$ or $n(n + 1)$ - both fail at line 2 (3 and 6 rather than 4).

Takeaway Verify a conjectured pattern against **every** given line, and check each component (start, length) separately.

Question 16 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from 1, 2, ..., 20. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up {1, 11}, {2, 12}, ..., {10, 20}: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.

Full method - how and why

A pigeonhole question. Build the 'boxes' so that two items in one box give exactly what you want, then answer boxes + 1.

- 1 Pair the numbers so that each pair differs by 10: {1, 11}, {2, 12}, ..., {10, 20}.

why: Every number from 1 to 20 lies in exactly one pair, and two numbers differ by 10 precisely when they share a pair.

- 2 There are 10 pairs, so 10 numbers can be chosen with no two in the same pair - for example 1 to 10.

why: This shows 10 is **not** enough, which is the half of the argument people skip.

- 3 With 11 numbers and only 10 pairs, two must come from the same pair - and those differ by 10.

why: Pigeonhole: more items than boxes forces a repeat.

- 4 So the smallest guaranteed n is 11.

why: Boxes + 1.

Where it goes wrong

- Answering 10 - the largest selection that can **avoid** the condition, not the smallest that forces it.
- Answering 2, confusing 'could happen' with 'guaranteed'.
- Forgetting to show 10 is achievable; without it you have not proved 11 is smallest.

Takeaway 'Smallest n that guarantees' = (largest bad selection) + 1. Construct the boxes so a shared box means success.

Question 17 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Full method - how and why

To disprove a claim about a test for inflexion, produce a function satisfying the conditions but failing the conclusion.

- 1 Need $f'(a) = f''(a) = 0$ with **no** inflexion at a .

why: Both hypotheses must hold, and the conclusion must fail.

- 2 Take $f(x) = x^4$ at $a = 0$: $f'(x) = 4x^3$ so $f'(0) = 0$ ✓; $f''(x) = 12x^2$ so $f''(0) = 0$ ✓.

why: Both derivatives vanish - the hypotheses are satisfied.

- 3 But x^4 has a **minimum** at 0 (it is positive on both sides), not a point of inflexion.

why: An inflexion requires the concavity to change sign; $f'' = 12x^2 \geq 0$ on both sides, so it does not.

- 4 Check the others: x^3 does have an inflexion at 0 (so it confirms, not refutes); x^2 has $f''(0) = 2 \neq 0$; $x^3 - 3x$ at $a = 1$ has $f'(1) = 0$ but $f''(1) = 6 \neq 0$; $f(x) = x$ has $f'(0) = 1 \neq 0$.

why: Each fails one of the two hypotheses, or satisfies the conclusion.

Where it goes wrong

- Choosing x^3 , the standard **example** of the claim rather than a counterexample.
- Choosing a function where one of the two derivative conditions fails at the stated point.

Takeaway $f''(a) = 0$ does not guarantee an inflexion; the second derivative must **change sign**. x^4 at 0 is the standard counterexample.

Question 18 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Full method - how and why

Ordering Euclid's proof. Again the skeleton is assume-deduce-contradict, and the dependencies fix the order.

- 1 II first: assume finitely many primes $p_1, \dots, p_n$ and construct $N = p_1 p_2 \cdots p_n + 1$.
why: Everything else refers to N , so it must be defined first.
- 2 Then I: each p_i divides $p_1 \cdots p_n$, so dividing N by p_i leaves remainder 1 - no listed prime divides N .
why: This is the key property of the $+1$.
- 3 Then IV: $N > 1$, so it has at least one prime factor.
why: Every integer greater than 1 has a prime factor - a separate fact, independent of I.
- 4 Finally III: that prime factor is not in the list, contradicting the assumption that the list was complete. Order: II, I, IV, III.
why: The contradiction needs **both** I (none of the listed primes works) and IV (some prime does).

Where it goes wrong

- Putting III immediately after II, before either supporting fact has been established.
- Thinking the proof claims N is prime - it need not be; it merely has a prime factor outside the list.

Takeaway In a contradiction proof, the contradiction is the last line, and it must rest on everything before it.

Question 19 BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I** $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1 + 1)^n$ expanded. All true.

Full method - how and why

Three standard properties of binomial coefficients. Each has a one-line reason and a one-line check.

- I:** $\binom{n}{r} = \binom{n}{n-r}$ - choosing r objects is the same as choosing the $n - r$ to leave. **True.**
why: Visible as the left-right symmetry of each row of Pascal's triangle.
- II:** $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ - Pascal's rule, i.e. each entry is the sum of the two above it. **True.**
why: Check with numbers: $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4 = \binom{4}{3}$ ✓.
- III:** $\sum_{r=0}^n \binom{n}{r} = 2^n$ - put $x = 1$ in $(1 + x)^n$. **True.**
why: Equivalently, a set with n elements has 2^n subsets, counted by size.
- 4 All three: 'I, II and III'.**
why: Match to the options.

Where it goes wrong

- Doubting Pascal's rule because of the index shift on the right - test it with small numbers.
- Thinking the sum in III is 2^{n+1} by miscounting the $n + 1$ terms.

Takeaway Symmetry, Pascal's rule and row sum 2^n are the three facts about $\binom{n}{r}$ worth knowing cold.

Question 20 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.

II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.

III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).

Full method - how and why

A subtle one: the **claim is true** but one line asserts more than was proved. Distinguish a false conclusion from an unjustified step.

1 Check line I: $f'(x) = 3x^2 - 6x + 3 = 3(x^2 - 2x + 1) = 3(x - 1)^2$. Correct.

why: Differentiate term by term, then factorise.

2 Check line II: a square is non-negative, so $f'(x) \geq 0$, with equality only at $x = 1$. Correct.

why: The word 'only' is right - $(x - 1)^2 = 0$ just at $x = 1$.

3 Line III says ' $f'(x) > 0$ for all x ' - **false at $x = 1$ **, where $f'(1) = 0$. So line III overstates line II.

why: It asserts a strict inequality that was never established.

4 Is the claim itself true? Yes: $f' \geq 0$ everywhere with equality at a single isolated point still gives $f(a) < f(b)$ for $a < b$.

why: A derivative vanishing at one point does not create a flat stretch; there is no turning point, because f' does not change sign.

Where it goes wrong

- Concluding the claim is false because $f'(1) = 0$ - a stationary point of inflexion is not a turning point.
- Blaming line I or II, both of which are correct.
- Calling the proof completely correct, ignoring the strictness error in line III.

Takeaway $f' \geq 0$ with equality only at isolated points still means strictly increasing - but a proof must not claim strict positivity it has not shown.